



Causal-CCM: A Personalized Causal Inference Framework for Disease Grading and Clinical Metric Regression in Corneal Confocal Microscopy

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ARTICLE INFO

Article history:

Oculomics, Causal Inference, Multimodal Learning, Corneal Confocal Microscopy, Diabetic Neuropathy, Digital Biomarkers, Medical AI.

ABSTRACT

The corneal sub-basal nerve plexus provides a sensitive, non-invasive window into systemic neuro-immune homeostasis, giving rise to the emerging field of “Oculomics.” However, the translation of Corneal Confocal Microscopy (CCM) into scalable screening is limited by scarce annotations, acquisition artifacts, and heterogeneous clinical tasks including grading, regression, and prognosis. Here, we present **Causal-CCM**, a causality-guided multimodal framework designed to reduce reliance on spurious image correlations while incorporating clinical context. The framework combines structural alignment, kernel-based feature screening, text-guided fusion, and a hyper-dynamic prediction head for joint disease grading and clinical metric regression. Across internal and external cohorts, Causal-CCM improves diagnostic and regression performance relative to strong deep learning baselines and shows improved robustness under simulated acquisition degradation. Frozen-representation transfer experiments on Diabetic Peripheral Neuropathy (DPN) and Herpes Simplex Keratitis (HSK) further suggest that the learned CCM features capture transferable neuro-immune morphology beyond the primary training tasks. These findings support Causal-CCM as a candidate digital surrogate for selected invasive or contact-based clinical measurements, while motivating prospective validation before broad deployment.

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1. Introduction

The cornea provides an accessible window into neural integrity, immune activity, and systemic metabolic injury. Corneal confocal microscopy (CCM) enables non-invasive, cellular-resolution visualization of the sub-basal nerve plexus and resident immune cells, offering quantitative information that is difficult to obtain through routine slit-lamp examination or conventional neurophysiological testing (Gad et al., 2022;

Alam et al., 2022). This neuroimmune information is clinically relevant to both ocular and systemic disorders. Dry eye disease (DED) involves interacting abnormalities in tear-film homeostasis, epithelial integrity, inflammation, and neurosensory regulation (Craig et al., 2017; Bron et al., 2017), whereas neuropathic corneal pain (NCP) arises from injury or dysfunction of corneal somatosensory pathways. Despite their different mechanisms and management strategies, DED and NCP frequently present with overlapping symptoms, including dryness, burning, photophobia, discomfort, and persistent ocular pain. NCP may additionally exhibit subtle nerve abnormalities, including microneuromas and aberrant regeneration, but these features are difficult to identify consistently and are not invari-

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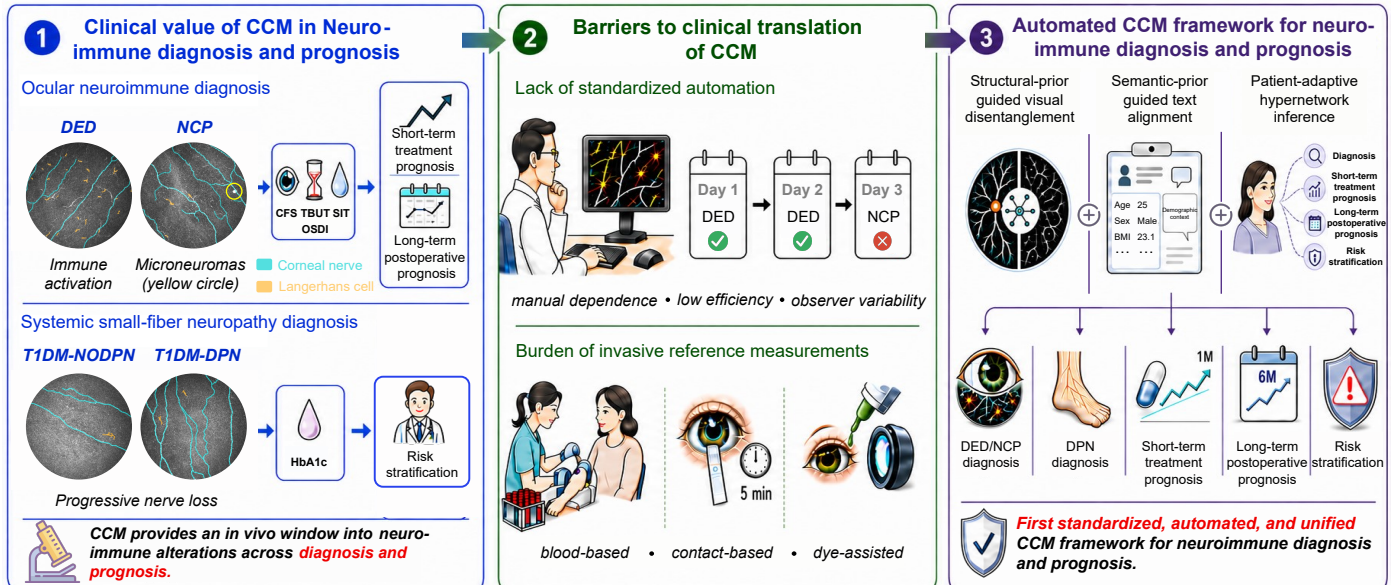


Fig. 1. From CCM-based neuroimmune phenotyping to automated diagnosis and prognosis. CCM reveals ocular neuroimmune and systemic small-fibre alterations, but clinical translation is limited by non-standardized manual assessment and burdensome reference measurements. The proposed framework integrates structural priors, clinical semantics, and patient-adaptive inference for disease diagnosis, short- and long-term prognosis, and risk stratification.

ably present (Dieckmann et al., 2017; Moein et al., 2020). CCM therefore offers an opportunity to distinguish clinically similar ocular neuroimmune disorders through their microscopic phenotypes rather than symptoms alone.

The same imaging modality can also characterize systemic small-fibre neuropathy. Diabetic peripheral neuropathy (DPN) is a progressive complication of diabetes that may lead to neuropathic pain, loss of protective sensation, foot ulceration, and lower-limb amputation (Pop-Busui et al., 2017). Its earliest pathological abnormalities frequently involve small nerve fibres, which may be insufficiently characterized by conventional nerve-conduction studies. Although skin biopsy allows direct quantification of intraepidermal nerve fibres, it requires tissue removal and specialist processing. By contrast, CCM can non-invasively quantify corneal nerve fibre loss, branching abnormalities, tortuosity, and regeneration, and these measurements have demonstrated diagnostic and longitudinal relevance to DPN (Gad et al., 2022; Perkins et al., 2021). HbA1c, when interpreted jointly with the corneal nerve phenotype, supports neuro-metabolic risk stratification, progression monitoring, and early intervention (Braffett et al., 2020). These complementary ocular and systemic applications motivate a common clinical pathway in which CCM first discriminates difficult neuroimmune diseases and subsequently anticipates their outcomes and risks, as summarized in Fig. 1.

Reliable prognosis, however, cannot be represented adequately by disease labels or coarse severity categories alone. For ocular-surface disease, corneal fluorescein staining (CFS) quantifies epithelial injury, tear-film break-up time (TBUT) reflects tear-film instability, and the Schirmer I test (SIT) characterizes tear secretion. Their continuous values provide complementary reference endpoints for evaluating short-term treatment response and longer-term ocular-surface recovery (Wolffsohn et al., 2017). For diabetes-related neuropathy, HbA1c reflects sustained glycaemic exposure and provides metabolic

context for interpreting corneal nerve damage and future neuropathic risk (Braffett et al., 2020). Two major clinical barriers nevertheless limit this pathway. First, conventional CCM assessment remains strongly dependent on manual image selection and expert interpretation, resulting in low efficiency, observer variability, and limited standardization (Alam et al., 2022). Second, the relevant clinical reference endpoints are obtained through separate contact-based, dye-assisted, or blood-based procedures. In addition to their invasive or burdensome nature, this fragmentation across image interpretation, ocular-surface testing, laboratory measurement, and longitudinal follow-up limits the scalability of CCM-based neuroimmune assessment.

Recent deep-learning studies have predominantly focused on corneal nerve and Langerhans-cell segmentation, automated nerve-tortuosity grading, and NCP diagnosis (Li et al., 2025b,c,a; Mou et al., 2022; Zhao et al., 2020; Koseoglu et al., 2025). A smaller parallel literature has investigated DPN classification as an isolated image-classification problem (Preston et al., 2022; Chen et al., 2024b). These studies demonstrate the feasibility of automated CCM analysis, but remain organized around individual anatomical structures or single clinical objectives. To the best of our knowledge, no previous automated CCM study has directly regressed any of the four clinically established continuous endpoints considered here—CFS, TBUT, SIT, or HbA1c—and no existing framework integrates difficult DED–NCP discrimination, diabetes-related neuropathy assessment, continuous clinical-endpoint estimation, short-term treatment prognosis, long-term postoperative prognosis, and risk stratification. Consequently, current systems neither provide a unified patient-level neuroimmune profile nor replace the fragmented collection of task-specific models and reference measurements. A conventional hard-sharing multi-task network is also insufficient: the objectives span categorical diagnosis, continuous regression, longitudinal prognosis, and risk stratifica-

tion, and their heterogeneous evidence requirements may produce task interference when all endpoints are forced to share the same representation.

To address these limitations, we propose a unified prior-guided and patient-adaptive framework for multi-task CCM analysis. The framework progressively addresses three sources of difficulty. First, structural-prior guided visual disentanglement aligns visual dependencies with training-derived clinical structure and screens task-relevant, nuisance-robust visual factors. Second, semantic-prior guided text alignment encodes available patient records as clinical semantic priors and uses cross-modal attention to contextualize visually ambiguous neuroimmune phenotypes. Third, patient-adaptive hypernetwork inference jointly conditions on patient context and task identity to generate prediction parameters for heterogeneous diagnostic, prognostic, regression, and risk stratification objectives. Direct prediction targets and information unavailable at the intended prediction time are excluded from the corresponding clinical inputs to prevent target leakage.

The main contributions of this work are threefold:

- **A unified automated clinical pathway for neuroimmune disease discrimination and outcome assessment.** To our knowledge, this is the first automated CCM framework that jointly addresses difficult ocular and systemic neuroimmune disease discrimination together with short-term treatment prognosis, long-term postoperative prognosis, and risk stratification within a single architecture. It automates the clinically challenging differentiation of DED, NCP, and controls, while also distinguishing controls, individuals with type 1 diabetes without DPN, and individuals with type 1 diabetes with DPN. This unified assessment can reduce phenotype-driven diagnostic ambiguity, support selection of disease-appropriate management, and enable earlier neuropathy-oriented surveillance.
- **Non-invasive estimation of burdensome clinical reference endpoints.** The framework directly regresses CFS, TBUT, SIT, and HbA1c from CCM images, replacing separate contact-based, dye-assisted, or blood-based measurements with continuous image-derived estimates. These quantitative endpoints support objective assessment of short-term treatment prognosis and long-term postoperative recovery for ocular-surface disease, while the joint interpretation of HbA1c and corneal nerve phenotypes supports neuro-metabolic risk stratification, progression monitoring, and early intervention.
- **A general prior-guided and patient-adaptive learning strategy.** We integrate structural-prior-guided visual disentanglement, semantic-prior-guided cross-modal alignment, and patient-adaptive hypernetwork inference to preserve clinically supported visual information while adapting the final decision function to each patient and task. This design provides a transferable learning paradigm for heterogeneous multi-task medical imaging.

2. Methods

As illustrated in Fig. 2, we propose a unified prior-guided and patient-adaptive framework for robust multi-task analysis of corneal confocal microscopy (CCM) images. The framework is organized into three complementary components that progressively address visual instability, semantic ambiguity, and inter-patient heterogeneity. First, structural-prior guided visual disentanglement constrains visual representation learning using training-derived clinical structure and separates pathology-relevant representations from nuisance-associated visual factors. Second, semantic-prior guided text alignment encodes patient clinical records as semantic priors and aligns them with the retained visual representations through cross-modal attention, thereby contextualizing visually ambiguous phenotypes according to clinical information. Third, patient-adaptive hypernetwork inference jointly conditions on patient semantics and task identity to dynamically generate prediction parameters for heterogeneous clinical objectives, including diagnosis, short-term prognosis, long-term prognosis, and risk stratification.

2.1. Structural-prior-guided visual disentanglement

CCM images contain clinically informative neural and immune patterns, while their visual representations may also be affected by acquisition conditions, background reflectivity, image quality, and cohort-specific correlations. A conventional image encoder optimized only by downstream prediction objectives may therefore retain visually predictive but clinically unstable information. We address this problem using structural-prior-guided visual disentanglement, which consists of three successive steps: visual dependency modeling, structural-prior alignment, and robust factor screening.

2.1.1. Visual representation and dependency modeling

For the s -th CCM image I_s , a visual encoder E_{vis} extracts a convolutional feature tensor $F_s = E_{\text{vis}}(I_s) \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of feature channels and $H \times W$ denotes the spatial resolution of the final convolutional representation. Let $F_s^i \in \mathbb{R}^{H \times W}$ denote the activation map of the i -th channel. To reduce scale differences among channels, each activation map is normalized to $[0, 1]$ as $\bar{F}_s^i = (F_s^i - \min(F_s^i)) / (\max(F_s^i) - \min(F_s^i) + \epsilon)$, where ϵ is a small constant for numerical stability.

We subsequently construct a visual dependency matrix $M_{\text{vis}} \in \mathbb{R}^{C \times C}$ based on normalized spatial co-activation. For channels i and j ,

$$M_{\text{vis}}(i, j) = \frac{1}{|\mathcal{B}|} \sum_{s \in \mathcal{B}} \frac{\sum_{p=1}^{HW} \bar{F}_{s,p}^i \bar{F}_{s,p}^j}{\sqrt{\sum_{p=1}^{HW} (\bar{F}_{s,p}^i)^2} \sqrt{\sum_{p=1}^{HW} (\bar{F}_{s,p}^j)^2} + \epsilon}, \quad (1)$$

where $\mathcal{B}$ denotes the current mini-batch and p indexes spatial positions. This normalized spatial-overlap formulation measures the degree to which two latent visual factors are activated over corresponding image regions while reducing the influence of absolute activation magnitude. The resulting matrix summarizes the pairwise topology of visual representations and provides the basis for subsequent alignment with clinical structural information.

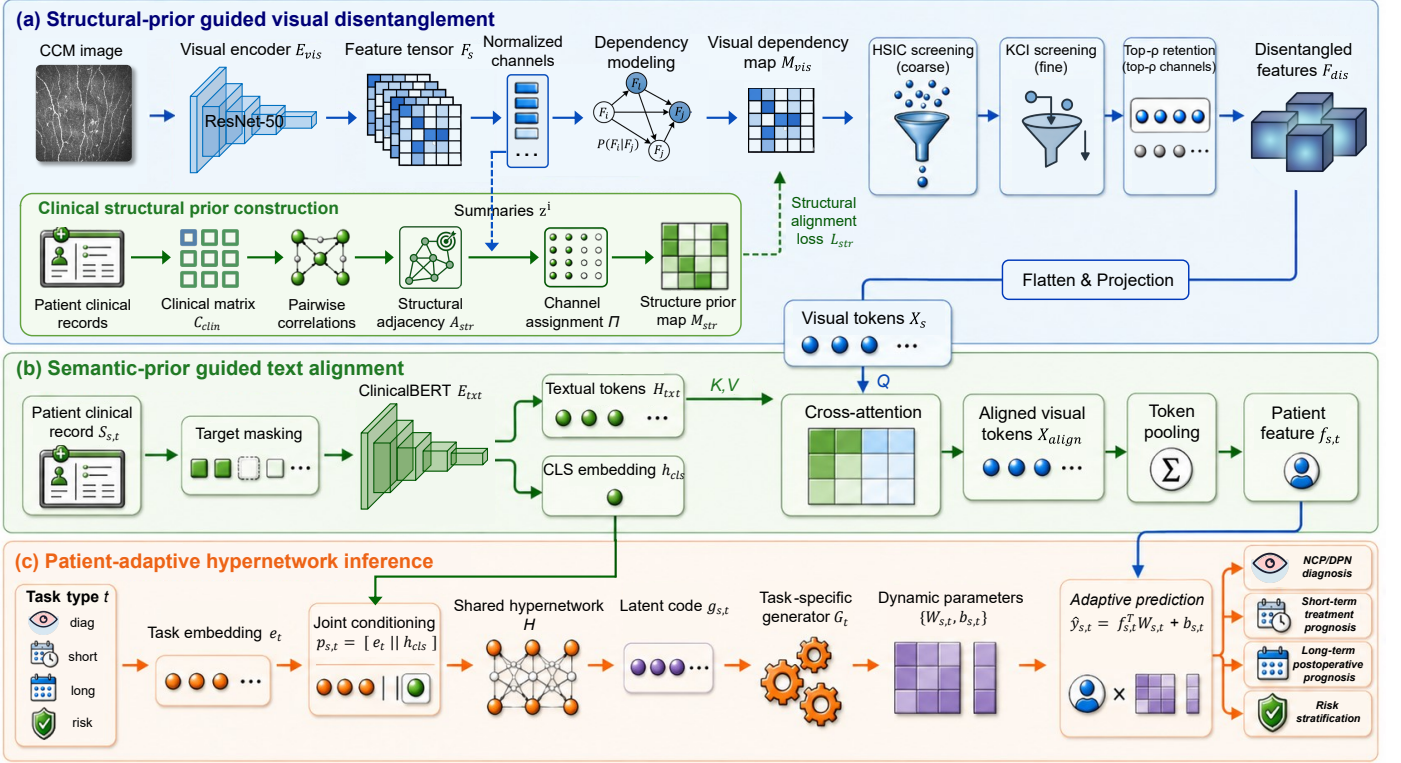


Fig. 2. Overview of the proposed framework. (A) Structural-Prior-Guided Visual Disentanglement uses clinically derived structural relationships to suppress unstable visual associations and retain pathology-relevant representations. (B) Semantic-Prior-Guided Text Alignment encodes patient clinical records as semantic priors and aligns them with visual representations to resolve phenotype ambiguity. (C) Patient-Adaptive Hypernetwork Inference dynamically generates patient- and task-specific prediction parameters for diagnosis, short-term prognosis, long-term prognosis, and risk stratification.

2.1.2. Clinical structural prior construction

Visual relationships estimated exclusively from imaging data may contain cohort- or acquisition-specific associations. We therefore construct a clinical structural prior using variables available in the training partition. Let $C_{\text{clin}} = [c^1, \dots, c^M] \in \mathbb{R}^{N \times M}$ denote the clinical matrix containing M available non-target variables from N training samples. Continuous variables are standardized using statistics estimated from the training partition. Binary variables are numerically encoded before association estimation. Variables that serve as direct prediction targets are excluded from the structural-prior variables.

A clinical structural adjacency matrix $A_{\text{str}} \in \mathbb{R}^{M \times M}$ is constructed according to pairwise relationships among the clinical variables as $A_{\text{str}}(m, r) = |\text{Corr}(c^m, c^r)|$, where $\text{Corr}(\cdot, \cdot)$ denotes Pearson correlation. For a binary variable, the same formulation corresponds to the point-biserial correlation after binary encoding. Because the clinical structural space and visual feature space have different dimensionalities, we establish a channel-to-clinical mapping. After an initial task-supervised visual warm-up, each visual channel is summarized across samples using global average pooling as $z_s^i = \text{GAP}(F_s^i)$. The association between visual channel i and clinical variable m is then estimated as $R(i, m) = |\text{Corr}(z_s^i, c^m)|$. The resulting channel-to-clinical association matrix is normalized across the clinical dimension as $\Pi(i, m) = \exp(R(i, m)/\tau) / \sum_{r=1}^M \exp(R(i, r)/\tau)$, where τ is a temperature parameter controlling assignment concentration.

The clinical structure is then projected into the visual-

channel space:

$$M_{\text{str}} = \Pi A_{\text{str}} \Pi^T \in \mathbb{R}^{C \times C}. \quad (2)$$

The channel-to-clinical mapping and the resulting structural prior are estimated exclusively from the training partition and are fixed during subsequent structural alignment.

2.1.3. Structural-prior alignment

The diagonal elements of both dependency matrices correspond primarily to self-association and are therefore removed before structural comparison. We define $O(M) = M - \text{Diag}(\text{diag}(M))$, and normalize the off-diagonal structures as $\widehat{M}_{\text{vis}} = O(M_{\text{vis}}) / (\|O(M_{\text{vis}})\|_F + \epsilon)$ and $\widehat{M}_{\text{str}} = O(M_{\text{str}}) / (\|O(M_{\text{str}})\|_F + \epsilon)$. The structural alignment loss is formulated as

$$\mathcal{L}_{\text{str}} = \frac{1}{C^2} \left\| \widehat{M}_{\text{vis}} - \text{sg}(\widehat{M}_{\text{str}}) \right\|_F^2, \quad (3)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $\text{sg}(\cdot)$ denotes the stop-gradient operator. This objective encourages visual representations to preserve clinically supported inter-feature structure while suppressing relationships inconsistent with the clinical structural prior construction.

2.1.4. Two-stage robust factor screening

Structural alignment regularizes the organization of the visual representation but does not explicitly remove individual

channels with weak task relevance or strong dependence on nuisance variables. We therefore introduce a two-stage kernel-based screening strategy. For sample s and channel i , a compact channel descriptor is defined as $x_s^i = [\text{GAP}(F_s^i), \text{GMP}(F_s^i)]$, where GMP denotes global maximum pooling. Therefore, $X^i = \{x_1^i, \dots, x_N^i\}$ contains one channel-level observation for each training sample.

Stage I: target-relevance screening via HSIC. We first evaluate the statistical dependence between each visual channel X^i and the downstream target Y using the Hilbert–Schmidt Independence Criterion (HSIC) (Gretton *et al.*, 2005). For continuous visual descriptors, a Gaussian radial-basis-function kernel is used, $K_i(s, r) = \exp(-\|x_s^i - x_r^i\|_2^2 / (2\sigma_i^2))$, where σ_i denotes the kernel bandwidth. For categorical targets, the target kernel is defined using a label kernel, $K_Y(s, r) = \mathbb{I}(y_s = y_r)$, whereas continuous targets are represented using an RBF kernel of the same general form. Let $H = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ denote the centering matrix. The centered kernels are $\tilde{K}_i = HK_iH$ and $\tilde{K}_Y = HK_YH$. The empirical HSIC statistic is then

$$T_{\text{HSIC}}^i = \frac{1}{(N-1)^2} \text{tr}(\tilde{K}_i \tilde{K}_Y). \quad (4)$$

Statistical significance is estimated using a permutation-based null distribution. For B random permutations of the target, the corresponding permutation p -value is calculated from the proportion of permuted HSIC statistics no smaller than the observed statistic. The resulting channel-wise p -values are corrected for multiple comparisons using the Benjamini–Hochberg procedure. Channels satisfying the predefined false-discovery-rate threshold are retained as target-relevant candidates.

Stage II: conditional robustness screening via KCI. A target-relevant feature may still reflect associations explained by demographic, acquisition, or clinical nuisance factors. We therefore perform kernel conditional-independence analysis using the available non-target conditioning variables $Z = [z_1, \dots, z_N]^T$ following the Kernel-based Conditional Independence (KCI) framework (Zhang *et al.*, 2011). A centered kernel matrix $\tilde{K}_Z$ is constructed for Z . The kernel residualization operator is defined as $R_Z = I - \tilde{K}_Z(\tilde{K}_Z + \lambda_{\text{KCI}}I)^{-1}$, where λ_{KCI} is the kernel regularization coefficient. Following the conditional-kernel formulation, the joint kernel for channel i and the conditioning variables is constructed using the Hadamard product $K_{\tilde{X}^i} = K_i \odot K_Z$, where $\odot$ denotes element-wise multiplication. Its centered form is denoted as $\tilde{K}_{\tilde{X}^i}$. The residualized kernels are $K_{\tilde{X}^i|Z} = R_Z \tilde{K}_{\tilde{X}^i} R_Z$ and $K_{Y|Z} = R_Z \tilde{K}_Y R_Z$.

The conditional-dependence statistic is

$$T_{\text{KCI}}^i = \frac{1}{N} \text{tr}(K_{\tilde{X}^i|Z} K_{Y|Z}). \quad (5)$$

The corresponding significance level is estimated from the KCI null distribution, and multiple comparisons across candidate channels are corrected using the same false-discovery-rate procedure. Channels that remain significantly associated with the target after conditioning are retained. Let $\mathcal{S}$ denote the final set of selected channels. The disentangled visual representation

is $F_{\text{dis},s} = F_s[\mathcal{S}, :, :] \in \mathbb{R}^{C_r \times H \times W}$, where $C_r = |\mathcal{S}| \leq C$. When the number of significant channels exceeds the predefined feature budget, channels are ranked according to their conditional-dependence statistics and the top fraction ρ is retained.

2.2. Semantic-prior-guided text alignment

Even after structural-prior-guided visual disentanglement, similar CCM phenotypes may correspond to different clinical conditions or disease states. Neural tortuosity, immune-cell activation, altered reflectivity, and related microstructural patterns can therefore remain visually ambiguous when interpreted without patient context. We address this problem by introducing a semantic-prior-guided text alignment module that uses available patient clinical information to contextualize the retained visual representations.

2.2.1. Clinical semantic prior encoding

For patient s and downstream task t , structured clinical information and available symptom descriptions are serialized into a standardized clinical sentence $S_{s,t}$. The input may contain demographic characteristics, disease history, systemic conditions, ocular characteristics, and free-text clinical symptoms available at the intended prediction time. To prevent information leakage, the direct target variable corresponding to task t , together with variables unavailable at the intended inference time, is masked from $S_{s,t}$ during both training and inference.

A ClinicalBERT encoder E_{txt} transforms the clinical sentence into contextual token embeddings $H_{\text{txt},s,t} = E_{\text{txt}}(S_{s,t}) \in \mathbb{R}^{L \times d_t}$, where L denotes the token sequence length and d_t denotes the text embedding dimension. The token-level representation $H_{\text{txt},s,t}$ serves as the semantic prior for cross-modal alignment. The contextual representation associated with the [CLS] token, $h_{s,t}^{\text{cls}} \in \mathbb{R}^{d_t}$, provides a patient-level semantic summary for subsequent adaptive inference. Numeric clinical values are retained as literal values within a fixed textual template, allowing the text encoder to model them jointly with the surrounding clinical context.

2.2.2. Semantic-prior-guided cross-modal alignment

The disentangled visual tensor is converted into a sequence of spatial visual tokens $X_{\text{vis},s} = \text{Flatten}_{\text{spatial}}(F_{\text{dis},s})^T \in \mathbb{R}^{HW \times C_r}$. The visual tokens are projected into a common multimodal embedding space as $X_s = X_{\text{vis},s} W_{\text{in}} \in \mathbb{R}^{HW \times d_a}$, where $W_{\text{in}} \in \mathbb{R}^{C_r \times d_a}$ is a learnable projection matrix. Cross-modal alignment is then implemented using standard scaled dot-product attention, with the visual tokens serving as queries and the clinical semantic tokens serving as keys and values. The attended semantic information is incorporated into the visual representation through residual modulation and layer normalization.

This operation allows individual visual regions to selectively retrieve clinical semantic information according to their correspondence with the patient record. The resulting task-specific patient representation is obtained through average aggregation over the aligned spatial tokens, yielding $f_{s,t} \in \mathbb{R}^{d_a}$.

2.3. Patient-adaptive hypernetwork inference

The downstream clinical objectives considered by the framework are heterogeneous and include diagnostic classification, short-term prognostic prediction, long-term prognostic prediction, and risk stratification. Different objectives may require different decision functions and may rely on different components of the same patient representation. We therefore employ a patient-adaptive hypernetwork to dynamically generate task-specific prediction parameters according to both patient context and task identity.

2.3.1. Task-patient joint conditioning

Let $\mathcal{T} = \{\text{diag, short, long, risk}\}$ denote the four clinical task families. For each task $t \in \mathcal{T}$, a learnable task embedding $e_t \in \mathbb{R}^{d_e}$ encodes the requested clinical objective. The task embedding is concatenated with the task-specific patient semantic representation as $p_{s,t} = [e_t \parallel h_{s,t}^{\text{cls}}] \in \mathbb{R}^{d_e+d_t}$, where $\parallel$ denotes vector concatenation. The joint conditioning representation therefore incorporates both task identity and patient-specific clinical context.

2.3.2. Hypernetwork parameter generation

A shared hypernetwork trunk $\mathcal{H}$ maps the joint conditioning representation to a latent parameter code $g_{s,t} = \mathcal{H}(p_{s,t}) \in \mathbb{R}^{d_h}$. Because heterogeneous tasks may have different output dimensionalities, a lightweight task-specific parameter generator G_t maps the shared latent code to the dynamic prediction parameters:

$$[\text{vec}(W_{s,t}); b_{s,t}] = G_t(g_{s,t}), \quad (6)$$

where $W_{s,t} \in \mathbb{R}^{d_o \times o_t}$, $b_{s,t} \in \mathbb{R}^{o_t}$, and o_t denotes the output dimensionality of task t . The shared hypernetwork trunk captures transferable relationships across clinical objectives, whereas the task-specific parameter generators preserve the output structure required by each prediction task.

2.3.3. Patient-adaptive prediction

For patient s and task t , the final prediction is computed as $\hat{y}_{s,t} = f_{s,t}^T W_{s,t} + b_{s,t}$. For categorical tasks, standard Softmax is used for multi-class prediction, while the sigmoid function is used for binary outcomes. For continuous clinical targets, $\hat{y}_{s,t}$ directly represents the predicted value. Accordingly, the prediction parameters $\theta_{s,t} = \{W_{s,t}, b_{s,t}\}$ are simultaneously patient-specific and task-specific, enabling the final decision function to adapt to heterogeneous patient contexts and clinical objectives.

2.4. Training strategy and joint optimization

The framework is optimized in two stages. The first stage establishes structurally constrained visual representations and determines the disentangled feature subset. The second stage performs semantic alignment and patient-adaptive multi-task inference.

2.4.1. Stage I: structural visual representation learning

During an initial warm-up period, the visual encoder is optimized using task-supervised auxiliary prediction heads. Let $u_s = \text{GAP}(F_s) \in \mathbb{R}^C$ denote the global visual representation. For each available training task t , an auxiliary static prediction head $\mathcal{P}_t$ generates $\tilde{y}_{s,t} = \mathcal{P}_t(u_s)$. Let Ω denote the set of observed sample-task pairs in the training data. The auxiliary visual prediction objective $\mathcal{L}_{\text{vis}}$ is computed as the task-balanced average of the corresponding task-specific losses, where λ_t denotes the task-balancing coefficient. After the warm-up representation is established, the channel-to-clinical association matrix Π and the structural prior M_{str} are estimated from the training partition and fixed.

The visual encoder is then optimized using

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{vis}} + \lambda_{\text{str}} \mathcal{L}_{\text{str}}, \quad (7)$$

where λ_{str} controls the contribution of structural-prior alignment. After Stage I convergence, HSIC-based target-relevance screening and KCI-based conditional robustness screening are performed on the training partition. The resulting channel subset $\mathcal{S}$ is fixed for subsequent training.

2.4.2. Stage II: semantic alignment and hypernetwork learning

During Stage II, the structurally optimized visual encoder and selected channel subset are fixed. The ClinicalBERT encoder, cross-modal alignment module, shared hypernetwork trunk, and task-specific parameter generators are optimized using the downstream task objectives. For mini-batch $\mathcal{B}$, let $\Omega_{\mathcal{B}} = \{(s, t) : s \in \mathcal{B}, y_{s,t} \text{ is available}\}$ denote the observed sample-task pairs. The downstream multi-task objective $\mathcal{L}_{\text{task}}$ is computed as the task-balanced average of the corresponding task-specific losses over $\Omega_{\mathcal{B}}$. Standard cross-entropy loss is used for categorical outcomes, whereas squared-error loss is used for continuous outcomes. To stabilize the dynamically generated heads, we additionally regularize their parameter magnitude using the average squared Frobenius norm of $W_{s,t}$ and squared ℓ_2 norm of $b_{s,t}$ over the observed sample-task pairs.

The final Stage-II objective is

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{task}} + \lambda_{\text{hyper}} \mathcal{L}_{\text{hyper}}, \quad (8)$$

where λ_{hyper} controls the regularization strength of the generated prediction parameters.

2.5. Task-specific leakage control

All statistics required by the proposed framework are estimated exclusively from the training partition. These include clinical structural relationships, channel-to-clinical associations, feature normalization statistics, kernel bandwidths, HSIC statistics, KCI statistics, feature-selection thresholds, and task-specific optimization parameters. For each downstream task t , the direct prediction target is removed from the corresponding clinical sentence $S_{s,t}$ during both training and inference. Clinical variables unavailable at the intended prediction time are likewise excluded. Variables used to construct the shared clinical structural prior are restricted to non-target

characteristics available within the training partition. Validation data are used only for model and hyperparameter selection, while independent test data remain isolated from prior construction, visual factor screening, normalization, and model optimization.

2.6. Inference

At inference time, a CCM image is first processed by the structurally optimized visual encoder, and only the channels selected during Stage I are retained. For the requested clinical task t , the available patient record is converted into the corresponding target-masked clinical sentence and encoded by ClinicalBERT. The semantic-prior-guided alignment module contextualizes the selected visual features using the patient-specific textual representation. Finally, the task embedding and patient semantic representation are jointly processed by the hypernetwork to generate the prediction parameters $\{W_{s,t}, b_{s,t}\}$, which are applied to the aligned multimodal representation to obtain the final task-specific prediction.

3. Experiments

3.1. Datasets and experimental setup

This study included five complementary clinical objectives: ocular neuroimmune diagnosis, systemic small-fiber neuropathy diagnosis, clinical reference metric regression, short-term treatment prognosis, and long-term postoperative prognosis. Detailed cohort composition, geographic distribution, sample size, and target definitions are summarized in Fig. 3. All data partitions were performed at the patient level to avoid information leakage.

Ocular neuroimmune diagnosis. This task evaluated three-class discrimination among healthy controls (HC), dry-eye disease (DED), and neuropathic corneal pain (NCP). Independent internal-development and external-validation cohorts were used to assess both in-domain performance and cross-center generalization.

Systemic small-fiber neuropathy diagnosis. This task evaluated three-class discrimination among HC, T1DM without diabetic peripheral neuropathy (T1DM-NoDPN), and T1DM with diabetic peripheral neuropathy (T1DM-DPN). Independent development and external cohorts were used to assess whether CCM could distinguish progressive systemic small-fiber neuropathy phenotypes across centers.

Clinical reference metric regression. Five continuous clinical targets were estimated from CCM: glycated hemoglobin (HbA1c), corneal fluorescein staining (CFS), tear-film breakup time (TBUT), Schirmer I test (SIT), and Ocular Surface Disease Index (OSDI). HbA1c characterized systemic glycemic status, whereas CFS, TBUT, SIT, and OSDI provided complementary objective and symptom-related ocular-surface measurements. As a downstream clinical analysis, predicted HbA1c was integrated with the independently learned DPN phenotype for neuro-metabolic risk stratification, with reference risk labels determined by senior clinicians. This joint assessment enabled identification of concordant and discordant metabolic-neural

states, thereby assisting clinicians in finer-grained neuropathy risk assessment and individualized clinical management.

Short-term treatment-response prognosis and assessment. Baseline CCM was used to predict one-month treatment response, including future CFS, TBUT, SIT, and OSDI profiles and the corresponding short-term response phenotype. At the one-month follow-up, paired baseline and follow-up observations were further used for treatment-response assessment, characterizing concordant recovery, persistent disease, symptom-dominant discordance, and sign-dominant discordance. These response phenotypes were subsequently compared with independently learned HC/DED/NCP disease phenotypes for cross-task etiological interpretation. This analysis supported short-term response prediction, follow-up assessment, and identification of patients with potential neurosensory involvement.

Long-term postoperative recovery prognosis. Baseline CCM after FS-LASIK was used to predict the six-month postoperative recovery outcome, defined as favorable recovery or persistent dry eye according to clinician-determined follow-up labels. At six months, the observed recovery outcome was further compared with independently learned HC/DED/NCP disease phenotypes, with favorable recovery expected to correspond predominantly to an HC-like phenotype and persistent dry eye to retain DED-like and/or NCP-like characteristics. This baseline-to-follow-up analysis supported long-term recovery prediction and postoperative phenotype validation for risk-aware follow-up.

Evaluation protocol. All images were acquired using the Heidelberg Retina Tomograph III with the Rostock Cornea Module (HRT-III/RCM) and analyzed at a spatial resolution of 384×384 pixels. The study was approved by the appropriate institutional ethics committees of all participating hospitals and by the Institute of Biomedical Engineering, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences (20240320-0022R). Each participating center conducted the study under its respective local ethical approval or applicable ethical waiver. All images and associated metadata were anonymized before analysis, and all study procedures were conducted in accordance with the Declaration of Helsinki.

3.2. Implementation details

A ResNet-50 backbone pretrained on ImageNet was used as the visual encoder, and pretrained ClinicalBERT encoded task-specific clinical records. Training followed the two-stage strategy in Sec. 2.4: Stage I performed visual warm-up, structural-prior alignment, and HSIC/KCI-based channel screening, after which the selected visual channels were fixed; Stage II optimized semantic cross-modal alignment and patient-adaptive hypernetwork inference. AdamW was used with an initial learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , cosine annealing to 1×10^{-6} , batch size 16, and a maximum of 500 epochs with early stopping after 20 epochs without validation improvement. Based on the sensitivity analysis, λ_{str} was set to 1.0 and the retained-channel ratio ρ to 30%.

All experiments used strict patient-level partitioning. Internal evaluation employed five-fold cross-validation with approximate 6:2:2 training/validation/test splits within each fold, and

Task	Cohort	Location	Images	Patients	Key details
1  Ocular neuroimmune diagnosis	Internal cohort (Development)	China 3 centers	5,425 images	518 patients	Targets: HC 1,936 • DED 2,094 • NCP 1,395
	External cohort (Validation)	China 2 independent centers	6,122 images	678 patients	Targets: HC 1,958 • DED 2,188 • NCP 1,976
2  Systemic small-fiber neuropathy diagnosis	Internal cohort (Development)	United Kingdom 2 centers	3,832 images	616 patients	Targets: HC 929 • T1DM-NODPN 1,289 • T1DM-DPN 1,614
	External cohort (Validation)	China & Italy 2 centers	5,265 images	450 patients	Targets: HC 1,307 • T1DM-NODPN 1,685 • T1DM-DPN 2,273
3  Clinical reference metric regression	Internal cohort (Development)	China 1 center	5,758 images	97 patients	Target: HbA1c, CFS, TBUT, SIT 5,758
	External cohort (Validation)	China 1 independent center	3,056 images	52 patients	Target: HbA1c, CFS, TBUT, SIT 3,056
4  Short-term treatment prognosis	Internal cohort (Development)	China 1 center	3,518 images	39 patients	Targets: ΔCFS, ΔTBUT, ΔSIT 3,518
	External cohort (Validation)	China 1 independent center	1,027 images	11 patients	Targets: ΔCFS, ΔTBUT, ΔSIT 1,027
5  Long-term postoperative prognosis	Internal cohort (Development)	China 1 center	1,629 images	70 patients	Targets: Non-persistent 439 • Persistent dry eye 1,190
	External cohort (Validation)	China 2 independent centers	1,378 images	59 patients	Targets: Non-persistent 518 • Persistent dry eye 860

Fig. 3. Overview of datasets and clinical evaluation tasks.

performance was reported as mean $\pm$ standard deviation across test folds. All data-dependent statistics and screening parameters were estimated exclusively from the training partition of each fold. Direct prediction targets and outcome-revealing variables were excluded from the corresponding clinical inputs, while external cohorts remained fully independent and were evaluated only after model development. Statistical comparisons with competing methods used two-sided paired Wilcoxon signed-rank tests with Bonferroni correction, with corrected $p < 0.05$ considered significant.

3.3. Evaluation metrics

Classification performance was evaluated using accuracy (ACC) and area under the receiver operating characteristic curve (AUC). For multi-class tasks, AUC was computed in a one-vs-rest manner and macro-averaged across classes. Regression performance for the clinical indicators was assessed using root mean square error (RMSE) and mean absolute error (MAE).

3.4. Comparison methods

For the diagnostic tasks, we compared our framework with eight representative methods, including four general-purpose approaches QCNN (Lao et al., 2025), VFMLG (Lu et al., 2025), Denseformer-MoE (Ding et al., 2025), and KGT Zhao et al. (2026) and four CCM-specific diagnostic methods M4 (Zhao et al., 2020), DeepGrading (Mou et al., 2022), HMSViT (Zhang et al., 2025), and NCP (Koseoglu et al., 2025).

For the clinical metric regression tasks, eight representative regression methods were included: QCNN (Lao et al., 2025), CardiacCLIP (Du et al., 2025), MReg (Liu et al., 2025), MLEDL (Huang et al., 2025), SupReMix (Wu et al., 2025), Denseformer-MoE (Ding et al., 2025), RegScore (Grzeszczyk

Table 1. Quantitative comparison of grading performance on DED and DR tasks (ACC and AUC) against representative baseline methods. Note that (↑) indicates higher values are better.

Grading Tasks	Metrics	Dry Eye Disease grading					Diabetic Retinopathy grading				p-value
		Control	Mild	Mod.	Severe	Overall	Control	NPDR	PDR	Overall	
M4 (Zhao et al., 2020)	ACC	0.751	0.724	0.706	0.792	0.743	0.821	0.854	0.839	0.838	<.001
	AUC	0.837	0.812	0.763	0.788	0.800	0.856	0.873	0.862	0.864	
DeepGrading (Mou et al., 2022)	ACC	0.803	0.791	0.776	0.842	0.803	0.862	0.911	0.894	0.889	<.001
	AUC	0.894	0.885	0.813	0.831	0.856	0.908	0.932	0.916	0.919	
CLIP-DR (Yu et al., 2024)	ACC	0.792	0.768	0.781	0.823	0.791	0.855	0.902	0.871	0.876	<.001
	AUC	0.868	0.859	0.824	0.810	0.840	0.897	0.915	0.922	0.911	
AOR-DR (Yu et al., 2025)	ACC	0.767	0.783	0.752	0.806	0.777	0.843	0.889	0.876	0.869	<.001
	AUC	0.851	0.866	0.801	0.814	0.833	0.879	0.905	0.893	0.892	
CausalNet (Carloni et al., 2023)	ACC	0.784	0.755	0.769	0.811	0.780	0.836	0.872	0.858	0.855	<.001
	AUC	0.842	0.828	0.809	0.791	0.818	0.865	0.884	0.877	0.875	
PelvicNet (Chen et al., 2024a)	ACC	0.812	0.786	0.774	0.805	0.794	0.874	0.896	0.881	0.884	<.001
	AUC	0.873	0.854	0.836	0.848	0.853	0.888	0.912	0.899	0.900	
Causal-CCM (Ours)	ACC	0.829	0.811	0.798	0.858	0.824	0.892	0.933	0.925	0.917	-
	AUC	0.921	0.919	0.853	0.857	0.888	0.935	0.966	0.958	0.953	

Table 2. Quantitative comparison of multi-target regression performance (RMSE and MAE) against representative baseline methods. Note that (↓) indicates lower values are better.

Regression Tasks	Metrics	HbA1c	TBUT	OSDI	CFS	SIT	Overall	p-value
JMMLRC (Brand et al., 2019)	RMSE	3.860	5.824	23.110	2.945	12.881	9.724	<.001
	MAE	2.741	4.015	18.062	2.010	9.516	7.269	
MTERL (Wang et al., 2019)	RMSE	3.752	5.790	22.843	2.809	11.230	9.285	<.001
	MAE	2.501	3.864	17.810	1.953	8.420	6.910	
BSTNet (Sun and Kosmas, 2024)	RMSE	4.015	5.922	21.650	2.684	10.551	8.964	<.001
	MAE	2.850	4.108	16.920	1.831	7.824	6.707	
MF-FS (Azriel et al., 2024)	RMSE	3.528	5.431	22.106	2.470	11.082	8.923	<.001
	MAE	2.380	3.655	17.204	1.705	8.219	6.633	
HRINet (Zhao et al., 2024)	RMSE	3.391	5.084	21.436	2.215	10.457	8.517	<.001
	MAE	2.230	3.321	16.650	1.558	7.642	6.280	
TCL (Xiao et al., 2024)	RMSE	3.184	4.352	20.897	1.956	9.613	8.000	<.01
	MAE	1.954	3.017	16.142	1.349	7.415	5.975	
Causal-CCM (Ours)	RMSE	2.970	3.645	19.508	1.649	8.720	7.298	-
	MAE	1.722	2.656	15.887	1.074	7.253	5.718	

et al., 2025), and KGT Zhao et al. (2026). All competing methods were evaluated under the same data partitions and task-specific evaluation protocol for fair comparison.

./figure/Experiment1.pdf

Fig. 4. Comparative visualization of pathological feature localization. The yellow dashed boxes in the leftmost column indicate expert-annotated ground truth regions.

4. Experimental Results

4.1. Quantitative Analysis

Superior Stratification in Disease Grading. As evidenced in Table 1, Causal-CCM achieves the strongest overall performance among the evaluated baselines across both DED and DR cohorts. In DED grading, our framework achieves an overall accuracy of 0.824 and an AUC of 0.888, surpassing the top-performing competitor, DeepGrading, by margins of 2.1% and 3.2%, respectively. This performance extends to DR stratification, where Causal-CCM attains an overall accuracy of 0.917 and an AUC of 0.953. The consistent gains across ocular surface and retinal pathology tasks suggest that the proposed causal-guided mechanism helps isolate robust pathological features from noise and mitigates semantic ambiguity in conventional CNNs.

Validation of Non-Invasive Digital Surrogates. Table 2 evaluates the framework’s capacity to estimate systemic and ocular surface biomarkers from corneal visual features. Causal-CCM yields the lowest error rates across all five physiological metrics among the evaluated methods, outperforming both multi-task learners (HRINet, TCL) and regression frameworks (JMMLRC, MTERL). Notably, in predicting HbA1c, our model achieved an RMSE of 2.970, indicating that CCM features contain information related to systemic metabolic status. For ocular metrics such as TBUT and CFS, which typically require contact-based or dye-based examinations, Causal-CCM achieved RMSEs of 3.645 and 1.649, respectively. These results suggest that the hyper-dynamic head can encode continuous physiological variables related to ocular homeostasis, although clinical replacement of standard tests requires prospective validation.

Table 3. Ablation study evaluating the incremental impact of core architectural components on dry eye disease (DED) and diabetic retinopathy (DR) grading (ACC and AUC) alongside multi-target clinical metric regression (average RMSE and MAE).

Exp. ID	Three components			DED Grading		DR Grading		Regression	
	C-Rep	T-Fuse	H-Dyn	ACC (↑)	AUC (↑)	ACC (↑)	AUC (↑)	RMSE (↓)	MAE (↓)
#1	✗	✗	✗	0.758	0.809	0.821	0.854	9.914	7.892
#2	✓	✗	✗	0.783	0.846	0.864	0.891	8.435	6.681
#3	✓	✓	✗	0.809	0.871	0.887	0.924	7.783	6.195
#4	✓	✓	✓	0.824	0.888	0.917	0.953	7.298	5.718

4.2. Qualitative Analysis

To assess interpretability, we visualized Class Activation Maps across disease severities. As shown in Fig. 4, baseline models such as CLIP-DR and DeepGrading often exhibit diffuse attention over non-pathological background regions, whereas Causal-CCM focuses more consistently on clinically relevant corneal structures. For DR, the model attention shifts from the dense, linear nerve plexus of controls toward regions with increased tortuosity and reduced density in NPDR and PDR. For DED grading, the attention mechanism emphasizes Langerhans cell activation, nerve curling, and denervation patterns across severity stages. This stage-specific attention pattern supports, but does not by itself prove, that the model uses contextually meaningful morphological features.

4.3. Ablation Study

4.3.1. Ablation of Key Components

To quantify the individual efficacy of our architectural innovations, we performed a stepwise ablation study on the CCM-RegINT test set. We defined a baseline using a standard ResNet-50 backbone with static multitasking heads and incrementally integrated three core components: Causal-Guided Visual Representation and Factor Inference (C-Rep), Text-Guided Multimodal Feature Fusion (T-Fuse), and Hyper-Dynamic Head Generation with Semantic Guidance (H-Dyn). The sequential design tests whether each component provides additional benefit under a fixed training protocol.

Visual De-noising via C-Rep (Exp. #1 → #2). The integration of Causal-Guided Representation yields the most substantial initial performance boost, improving DED accuracy by 2.5% and reducing the regression MAE by 1.211. This gain is consistent with the hypothesis that raw CCM features contain non-pathological artifacts, such as illumination shifts, and that structural alignment with clinical priors helps reduce spurious correlations.

Semantic Disentanglement via T-Fuse (Exp. #2 → #3). Incorporating Text-Semantic Fusion improves complex pathology discrimination, specifically elevating DR grading accuracy by 2.3%. In the absence of clinical context, visual features of nerve tortuosity in DED and DR can overlap; textual priors, such as diabetes duration, provide semantic anchors that help refine decision boundaries.

Adaptive Inference via H-Dyn (Exp. #3 → #4). The final deployment of the Hyper-Dynamic Head provides the best overall performance, further reducing regression MAE to 5.718 and

Table 4. Sensitivity analysis of key hyperparameters and architectural strategies. We report (ACC and AUC) for grading, and (RMSE and MAE) for regression. The optimal settings are highlighted.

Hyperparameter Setting	Value (λ / ρ / Strategy)	DED Grading ACC AUC		DR Grading ACC AUC		Regression RMSE MAE	
(a) Weight of Structural Alignment Loss (λ_{map}):							
λ_{map}	0.1	0.786	0.851	0.874	0.915	8.243	6.559
	0.5	0.811	0.879	0.901	0.942	7.542	5.884
	1.0 (Ours)	0.824	0.888	0.917	0.953	7.298	5.718
	1.5	0.795	0.861	0.886	0.927	7.934	6.241
	2.0	0.769	0.838	0.859	0.898	8.691	6.912
(b) Retention Ratio of Causal Features (ρ_{causal}):							
ρ_{causal}	10%	0.743	0.796	0.831	0.865	9.104	7.237
	30% (Ours)	0.824	0.888	0.917	0.953	7.298	5.718
	50%	0.816	0.881	0.909	0.946	7.462	5.851
	70%	0.791	0.853	0.879	0.919	8.053	6.382
	100% (Base)	0.758	0.809	0.821	0.854	9.914	7.892
(c) Impact of Clinical Context Integration Strategy:							
Clinical Integration	Structured Vector (MLP)	0.796	0.862	0.894	0.927	7.863	6.115
	Complaint Embedding (BERT)	0.782	0.849	0.883	0.914	8.141	6.438
	Causal-Guided Factor Inf. (Ours)	0.824	0.888	0.917	0.953	7.298	5.718
(d) Impact of Multi-Task Head Strategy:							
Head Strategy	Single-Task Specialists	0.819	0.897	0.905	0.941	7.558	5.742
	Static Multi-Task	0.794	0.853	0.881	0.918	8.462	6.671
	Hyper-Dynamic (Ours)	0.824	0.888	0.917	0.953	7.298	5.718

improving grading metrics. Unlike static heads that may suffer from negative transfer in heterogeneous multi-task learning, the hypernetwork generates patient-specific parameter spaces on-the-fly and helps balance the feature distributions required for discrete classification and continuous regression.

4.3.2. Hyperparameter and Mechanism Sensitivity Analysis

Regularization vs. Overfitting (λ_{map}). Table 4(a) reveals an empirical optimum at $\lambda_{map} = 1.0$. Lower weights ($\lambda = 0.1$) reduce the influence of structural alignment, degrading performance (RMSE $\uparrow$ 0.945). Conversely, excessive regularization ($\lambda = 2.0$) may force rigid adherence to population statistics, suppressing patient-specific visual nuances (DR accuracy $\downarrow$ 5.8%). This suggests that clinical knowledge should guide, not override, visual perception.

Signal-to-Noise Optimization (ρ_{causal}). As shown in Table 4(b), feature retention exhibits an empirical optimum. Retaining 100% of features is associated with higher regression RMSE (9.914), whereas aggressive pruning ($\rho = 10\%$) likely discards useful pathological details. The optimal performance at $\rho_{causal} \approx 30\%$ suggests that a substantial proportion of raw feature channels may be task-irrelevant or acquisition-sensitive, but it should not be interpreted as a direct estimate of non-causal image content.

4.3.3. Fine-grained Analysis

Efficiency of Causal-Driven Clinical Integration. Table 4(c) evaluates the efficacy of different clinical data utilization paradigms. We compared our proposed Causal-Guided Visual Representation and Factor Inference strategy against two prevalent fusion baselines: 1) *Structured Vector (MLP)*, where discrete clinical variables (e.g., HbA1c, duration) are concatenated into a vector, projected via an MLP, and fused with visual features; 2) *Complaint Embedding (ClinicalBERT)*, where the patient’s unstructured chief complaint report is encoded directly by ClinicalBERT and fused.

./figure/Experiment2.pdf

Fig. 5. t-SNE visualization of latent feature manifolds across grading tasks.

Results demonstrate that our causal-guided strategy outperforms both baselines (e.g., reducing regression RMSE by 0.565 compared with MLP). While the MLP baseline captures raw values, it lacks semantic context; conversely, the complaint embedding captures semantics but misses precise quantitative physiological constraints. Our method uses clinical priors to structure the visual dependency map and infer latent factors, ensuring that visual features are not merely concatenated with clinical data but are regularized by clinically plausible relationships.

Efficiency of Hyper-Dynamic Head Generation. Table 4(d) quantifies the impact of our Hyper-Dynamic mechanism against a static multi-task baseline and an ensemble of single-task specialists. The static baseline shows performance degradation (DR accuracy $\downarrow$ 3.6%), consistent with negative transfer between discrete grading and continuous regression. In contrast, the Hyper-Dynamic approach reduces this conflict and achieves performance competitive with single-task specialists while retaining a unified model.

4.3.4. Visualization of Latent Feature Manifolds

To visualize discriminative structure, we projected the high-dimensional feature spaces of the test set onto two dimensions using t-SNE. Fig. 5 contrasts the manifold topology of Causal-CCM against baselines. A persistent challenge across both DED and DR grading is the ambiguous boundary between intermediate pathologies; baselines show greater overlap where *Mild/Moderate* DED and *NPDR* samples remain entangled with controls. By enforcing structural constraints to reduce spurious environmental noise, Causal-CCM yields a more organized manifold with lower apparent intra-class variance and larger inter-class margins. This visualization is qualitative but consistent with the quantitative gains.

5. Discussion and Further Analysis

5.1. Short-term treatment-response prognosis

Response phenotype classification. We first evaluated whether CCM-derived longitudinal clinical measurements could reconstruct clinically defined short-term treatment-response phenotypes. CFS, TBUT, and SIT were independently estimated from baseline and one-month CCM images, and their longitudinal changes were combined with symptom information to derive four response phenotypes: concordant recovery, persistent disease, symptom-dominant discordance, and sign-dominant discordance. Reference phenotype labels were determined from measured clinical indices, OSDI, and clinician-assessed one-month follow-up outcomes. The resulting confusion matrix achieved an overall accuracy of 91.00% (Fig. 6a). One-vs-rest ROC analysis yielded AUCs of 0.92, 0.89, 0.94, and 0.91 for concordant recovery, persistent disease, symptom-dominant discordance, and sign-dominant discordance, respectively (Fig. 6b). These results indicate that CCM-derived longitudinal biomarkers can reliably identify clinically meaningful short-term treatment-response phenotypes.

Cross-task phenotype concordance and clinical decision support. We next applied the independently trained HC/DED/NCP diagnostic model to the same one-month CCM images. Concordant recovery was predominantly HC-like (90.71%), persistent disease was predominantly DED-like (88.58%), symptom-dominant discordance was predominantly NCP-like (94.12%), and sign-dominant discordance was predominantly DED-like (92.98%), yielding an overall cross-task concordance of 91.59% (Fig. 6c). This correspondence linked longitudinal clinical-metric regression with independently learned disease representations and provided an etiology-oriented interpretation of treatment response. Based on these phenotypes, patients were further assigned to maintenance of current therapy, escalation of ocular-surface treatment, or alternative neurosensory evaluation. Compared with clinician-defined management decisions, the AUC was 0.95 for maintenance of current therapy, 0.92 for escalation of ocular-surface treatment, and 0.94 for neurosensory evaluation (Fig. 6d). Together, these findings show that the derived response phenotypes not only align with independently learned disease states but also provide clinically meaningful information to assist subsequent treatment decision-making.

5.2. Robustness Analysis: Invariance to Acquisition Degradation

To evaluate clinical utility under suboptimal imaging conditions, we subjected the model to a controlled “stress test” simulating realistic acquisition degradations, including Gaussian blur, speckle noise, illumination shifts, and contrast jitter. Fig. 7 quantifies the stability of our framework against baselines across both grading (accuracy) and regression (RMSE) tasks. Standard deep learning models such as DeepGrading and CLIP-DR showed performance decay under perturbation, with grading accuracy dropping by 6.1% ~ 8.5% and regression RMSE increasing by 0.983~1.231. This vulnerability is consistent with reliance on acquisition-sensitive pixel statistics. In

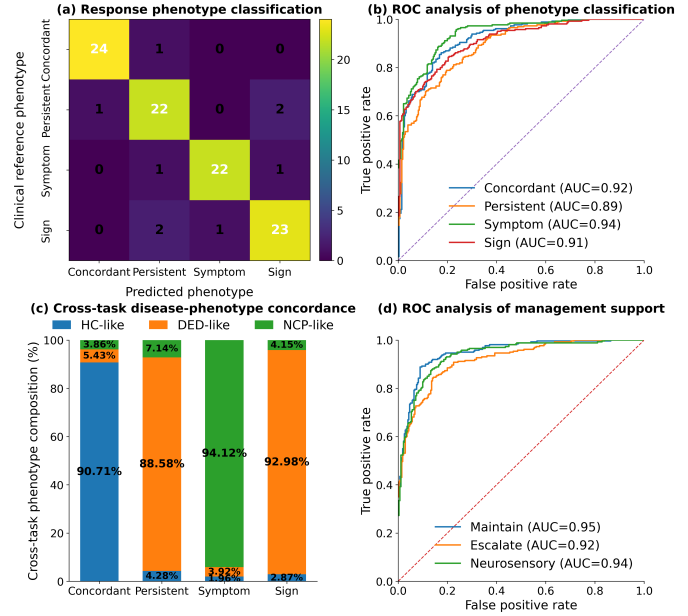


Fig. 6. Clinical validation of short-term treatment-response prognosis. a, Confusion matrix for classification of concordant recovery, persistent disease, symptom-dominant discordance, and sign-dominant discordance. **b,** One-vs-rest ROC curves for the four treatment-response phenotypes. **c,** Cross-task concordance between treatment-response phenotypes and independently predicted HC-like, DED-like, and NCP-like disease phenotypes. **d,** One-vs-rest ROC curves for phenotype-guided clinical management, including maintenance of current therapy, escalation of ocular-surface treatment, and neurosensory evaluation.

./figure/Discussion1.pdf

Fig. 7. Robustness analysis of discrete grading and continuous regression tasks under simulated image acquisition degradations.

contrast, Causal-CCM limited the accuracy decline to 3.0% and RMSE fluctuation to 0.583, supporting the role of HSIC and KCIT constraints in reducing sensitivity to background noise.

./figure/Discussion2.pdf

Fig. 8. External validation evaluating cross-center generalization performance on the CORN-Pro dataset.

5.3. External Generalization: Cross-Center Disease Grading

To validate the framework’s generalization capability across clinical settings, we performed direct inference on an external, multi-center DED dataset characterized by device-specific heterogeneity. Causal-CCM achieved an AUC of 0.827 (Fig. 8) and outperformed the strongest evaluated competitors, including PelvicNet (0.799) and CausalNet (0.792). These findings suggest that the unified causal mechanism extracts features that are more stable under cross-center shift, although broader prospective validation remains necessary.

5.4. External Generalization: Diagnosis of Unseen Pathologies

To verify whether Causal-CCM captures corneal biomarkers beyond the primary training tasks, we evaluated frozen-representation transfer on two unseen pathologies: Diabetic Peripheral Neuropathy (DPN) and Herpes Simplex Keratitis (HSK). To reduce semantic data leakage, pathology-specific keywords were masked in the ClinicalBERT inputs. We detached the hyper-dynamic head and used the frozen backbone to extract latent features for downstream linear-probe analysis.

Statistical Significance of Latent Representations. We first investigated the discriminative power of the extracted latent features relative to disease pathology. As visualized in Fig. 9a, for the DPN cohort, the features exhibit monotonic stratification across four severity groups (Control, SPK, NODPN, DPN), reflecting sensitivity to progressive neurodegeneration. Conversely, in the HSK cohort (Fig. 9b), feature distributions reveal a demarcation between controls and active infection ($p < 0.001$, independent t-tests), indicating responsiveness to inflammatory immune activation. These patterns suggest that the encoder captures morphological anomalies, such as nerve tortuosity and immune cell density, beyond the original task labels.

./figure/Discussion3.pdf

Fig. 9. Cross-disease generalization performance on unseen neural and inflammatory pathologies presenting violin plots of latent feature distributions across severity cohorts for DPN in panel (a) and HSK in panel (b) alongside comparative diagnostic AUC performance against a morphological multivariate baseline and SOTA methods in panels (c) and (d), respectively.

Diagnostic Performance Comparison. Subsequently, we benchmarked diagnostic precision using a linear Logistic Regression (LR) probe. We compared our latent representations against a “Multivariate” baseline, constructed by quantifying morphological parameters of corneal nerves and Langerhans cells (extracted by CCM-Pro (Li et al., 2025b) and MorphoBoost (Li et al., 2025a), respectively) and aggregating them via multivariate regression. As quantified in Fig. 9c, Causal-CCM secures the highest performance among the evaluated representations. In multi-class DPN severity grading, our method achieves a macro-averaged one-vs-rest AUC of 0.746, surpassing both the hand-crafted Multivariate baseline (0.716) and supervised deep learning competitors (e.g., CLIP-DR: 0.705). Similarly, for HSK diagnosis (Fig. 9d), the framework attains an AUC of 0.985, outperforming the Multivariate approach (0.953). These results indicate that the learned representations encode pathological cues that complement standard morphometry.

5.5. External Validation: Objective Metric Regression

Beyond static diagnosis, a useful digital surrogate should quantify physiological metrics and support disease trajectory estimation across unseen domains. We utilized the independent **CCM-RegEXT** dataset to evaluate the reliability of our framework against six multi-task baselines: JMMLRC, MTERL, BSTSNet, MT-FS, HRINet, and TCL.

To assess cross-center generalization, we evaluated the model’s performance in inferring continuous clinical indices (TBUT, CFS, SIT, and OSDI) on the external CCM-RegEXT cohort.

./figure/Discussion4.pdf

Fig. 10. External validation of objective clinical metric regression on the CCM-RegEXT cohort. (a) Comparative external regression performance. (b-e) Pearson correlation analysis.

- **Error Minimization:** As illustrated in Fig. 10a, Causal-CCM achieves the lowest external regression error with an RMSE of 12.972, outperforming the best-performing baseline TCL (13.578 on the external cohort). This supports the robustness of our method to heterogeneous data across different centers.
- **Correlation Analysis:** To further assess consistency between predictions and clinical reference measurements, we conducted Pearson correlation analysis. As visualized in Fig. 10(b-e), the model exhibits statistically significant positive correlations ($p < 0.001$) across all four indicators, achieving peak consistency in CFS ($r = 0.381$) and OSDI ($r = 0.368$).

The combination of lower regression error and significant correlation coefficients suggests that Causal-CCM partially mitigates domain shift. It may help map corneal nerve morphology to clinical severity without requiring physical contact or fluorescein staining, but replacement of standard clinical examinations remains outside the scope of this retrospective study.

5.6. Longitudinal Validation: Prognosis and Treatment Guidance

Leveraging the longitudinal structure of the CCM-RegEXT dataset ($T_0 \rightarrow T_1$), we evaluated whether baseline CCM features are associated with short-term treatment response. To establish ground truth, patients were stratified into “Responders” (defined by clinical improvement: OSDI reduction $> 30\%$ or TBUT increase $> 5s$) and “Non-Responders”. Based on this classification, we conducted a two-step analysis:

./figure/Discussion5.pdf

Fig. 11. Longitudinal evaluation of short-term treatment prognosis and treatment response guidance. (a-d) Violin plots showing the discriminative distribution of latent features between Non-Responders and Responders across four metrics. (e) Comparative prognostic efficacy assessment (AUC) against SOTA baselines.

- **Discriminative Power Analysis (Violin Plots):** First, we visualized the distribution of baseline latent features for patients stratified into “Responders” and “Non-Responders” across four clinical metrics (SIT, OSDI, CFS, TBUT). As shown in the violin plots (Fig. 11a-d), the framework reveals significant statistical differences ($p < 0.001$) in feature activation between the two groups. This indicates that the model captures morphological signals associated with treatment sensitivity.
- **Treatment Response Prediction:** Building on this separability, we performed binary classification to predict treatment outcomes using only baseline images. As quantified in Fig. 11e, Causal-CCM achieved an AUC of 0.742, outperforming baselines including CLIP-DR (0.721), PelvicNet (0.715), and DeepGrading (0.697). This suggests that the method may help identify patients more likely to respond to treatment, pending prospective validation.

5.7. Limitations

This study has several limitations. First, CCM-RegINT and CCM-RegEXT were collected retrospectively, and some patients had incomplete multi-task annotations; although we used patient-level splitting and masked target variables from task-specific inputs, prospective validation is required to exclude residual selection bias. Second, several external analyses use frozen representations and linear probes rather than fully independent prospective deployment, so they should be interpreted as transferability evidence rather than definitive zero-shot diagnosis. Third, the statistical analyses are constrained by heterogeneous label availability and cohort size, particularly for

longitudinal response prediction. Finally, the proposed digital surrogate should be viewed as a screening and triage aid until its calibration, failure modes, and clinical decision impact are evaluated in larger multi-center studies.

6. Conclusion

In this study, we introduce **Causal-CCM**, a unified framework that combines structural alignment, kernel-based feature screening, text-guided fusion, and hyper-dynamic inference for CCM-based disease grading and clinical metric regression. Across internal and external evaluations, the framework improves performance over representative baselines and shows reduced sensitivity to simulated acquisition degradation. The results support the oculomics hypothesis that corneal neuro-immune signatures contain information related to ocular and systemic disease status. Causal-CCM therefore provides a promising non-invasive screening and triage framework, while prospective multi-center validation is required before clinical replacement of standard examinations can be claimed.

7. Acknowledgement

This work is supported by the National Natural Science Foundation of China (62422122, 62272444, 62401551), Key research and development program of Zhejiang Province (2024C03204, 2023C04017, 2024C03101), the Project of Ningbo Public Welfare Science and Technology (2023S012, 2024S058), Ningbo 2025 S&T Mega project (2022Z134).

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